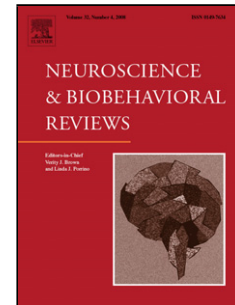


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## RDoC: The Case for ACC

The Research Domain Criteria Framework:

The Case for Anterior Cingulate Cortex

Running Head: RDOC: THE CASE FOR ACC

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## RDoC: The Case for ACC

## Highlights

- Anterior cingulate cortex contributes to individual differences in personality.
- The maladaptive expression of these traits contributes to multiple mental disorders.
- These relationships are revealed by electrophysiological signals produced in ACC.

## Abstract

The United States National Institute of Mental Health has recently promoted the Research Domain Criteria framework, which emphasizes the study of neurocognitive constructs that cut across different disorders. These constructs are said to express dimensionally across the population, giving rise to psychopathologies only in the extreme cases where that expression is maladaptive. Inspired by the RDoC framework, we propose that recent insights into the function of anterior cingulate cortex (ACC), a brain area said to be responsible for selecting and motivating extended behaviors, may elucidate the etiology of a diverse array of mental disorders. We argue that ACC function contributes to individual differences in personality traits related to reward sensitivity and persistence, and propose that the maladaptive expression of these traits contributes to multiple mental and neurological disorders. Our discussion is organized around a computational framework that relates the reward processing and control functions of ACC, as revealed by two electrophysiological phenomena called the reward positivity and frontal midline theta oscillations, to a distributed neural system underlying cognitive control.

## Keywords

anterior cingulate cortex, research domain criteria framework, persistence, reward sensitivity, depression, schizophrenia, obsessive compulsive disorder, Parkinson`s disease, attention-deficit hyperactivity disorder, substance abuse, reward positivity, frontal midline theta

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## The Challenge of Diagnosing Mental Disorders

Recent years have seen a growing appreciation that differences between normal mental health and abnormal mental impairment are ones of degree rather than kind. In a striking example of this shift, the British Psychological Society has recently stated that, “One of the most important messages of this report is that there is no dividing line between ‘psychosis’ and ‘normality’. There is no ‘us’ and ‘them’” (British Psychological Society, 2014). The Society emphasizes that hearing voices when no one else is present – once seen as the hallmark of schizophrenia – is commonly experienced by people who do not otherwise exhibit pathology nor necessarily find the experiences to be disturbing. On this account, the symptoms of schizophrenia are expressed along a continuum, with many people exhibiting none of the symptoms and other people expressing more; the symptoms only become clinically relevant in extreme cases when that expression is maladaptive.

Despite this evolving perspective, most existing systems for diagnosing mental disorders are polythetic: Individuals are diagnosed with a disorder when they exhibit more than a threshold number of symptoms. This approach is heuristically convenient but problematic for a number of reasons. In particular, because polythetic classification systems are dichotomous, they coarsely group individuals at intermediate levels of vulnerability for a disorder with other people who exhibit either all or none of the symptoms of the disorder. The approach also categorizes individuals with non-overlapping subsets of symptoms into the same group, which can result in individuals with starkly different clinical profiles sharing the same disorder. For instance, the

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Diagnostic Statistical Manual (DSM)-IV defines depression according to a threshold criterion of four out of nine possible depression-related symptoms, which yields a total of 227 different combinations of symptoms that meet the diagnosis; when these symptoms are further subdivided into related sub-components of depression, the grand total balloons to 1497 combinations (Østergaard, Jensen, & Bech, 2011). The polythetic approach also permits different disorders to share particular symptoms in common, which raises questions about their nosological status. For example, anxiety and depression often co-occur, suggesting that comorbid anxiety and depression should be considered as a separate disorder distinct from pure anxiety and pure depression, and that the pure forms of the disorders may not even exist (Schwartz, 2011).

Rising to this challenge, the United States National Institute of Mental Health (NIMH) has advocated for the study of biologically-based intermediate functions that elucidate the causal factors underlying disorder symptomatology (Cuthbert & Insel, 2013; Cuthbert & Kozak, 2013; Insel et al., 2010; Sanislow et al., 2010). This Research Domain Criteria (RDoC) framework promises to “develop, for research purposes, new ways of classifying mental disorders based on behavioral dimensions and neurobiological measures” that “starts with basic mechanisms and studies dysfunctions in these systems as a way to understand homogeneous symptom sets that cut across multiple disorders, rather than starting with clinical symptoms and working backwards” (National Institute of Mental Health, 2015). This framework conceives of mental disorders according to an array of underlying, causative “constructs”, as illustrated in Figure 1. The top panel indicates the conventional approach, where each element in the top row indicates a potential disorder as specified by the current diagnostic systems (such as depression, schizophrenia, anorexia, and so on), and each arrow represents a potential symptom; on this view, a specific disorder (dashed red box) is characterized by a cluster of symptoms (dashed red

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arrows). By contrast, the RDoC approach (bottom panel) emphasizes the set of underlying functional constructs, each of which is understood to be mediated by a specific neural system that is vulnerable to disruption (for example, “fear processing” or “visual perception”). From this perspective, disruption to a particular construct (red dashed box) gives rise to symptomatology that can contribute to multiple different disorders (red dashed arrows). Crucially, the RDoC approach holds that each of these constructs are expressed dimensionally across the population, from typical levels of expression (solid black arrows) to clinically-elevated levels (dashed red arrows).

Furthermore, the RDoC framework conceives of these constructs in terms of a multi-level matrix that decomposes high-level “domains” into constructs and sub-constructs. In turn, these constructs can be interrogated using an array of empirical “units” of analysis that target different neurocognitive levels of processing (namely, “Molecules,” “Cells,” “Circuits,” “Physiology,” “Behavioral measures,” and “Self-Reports”), together with specified “Task Paradigms” appropriate to the level of analysis. In order to provide a heuristic point of departure for further research into the issue, the contents of this matrix have been tentatively populated by a series of working groups convened by NIMH between 2010-2012. As of the time of this writing, five domains are proposed to comprise the top level of the matrix: “Negative Valence Systems,” “Positive Valence Systems,” “Cognitive Systems,” “Social Process Systems,” and “Arousal/Regulatory Systems,” each of which are said to comprise a set of constructs. For example, the Cognitive System includes the constructs “attention,” “perception,” “working memory,” “declarative memory,” “language behavior,” and “cognitive (effortful) control.” The scientist is then tasked with relating neurobehavioral measures – such as single unit activity in prefrontal cortex, the functional magnetic resonance imaging (fMRI) blood-oxygen-level

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dependent (BOLD) response in the hippocampus, performance on a particular task, and so on – to aspects of psychopathology, rather than to specific disorders as currently defined. This approach aligns well with previous efforts to identify endophenotypes, intermediate phenotypes, and biomarkers of psychopathology (Lenzenweger, 2013), as well as with formal computational approaches that illustrate how mental disorders can emerge from disruption to individual elements within complex neural networks (Adams, Huys, & Roiser, 2015; Maia & Frank, 2011; Montague, Dolan, Friston, & Dayan 2012).

Although we applaud this new emphasis on etiology over symptomatology, given that many of the proposed constructs are themselves poorly understood, we question whether populating the RDoC matrix in this manner has been premature. Anterior cingulate cortex (ACC) is a case in point. Table 1 lists all of the constructs and sub-constructs associated with units that have been attributed to ACC as of the time of this writing (National Institutes of Mental Health, 2015). These constructs are said to support a wide variety of functions underlying all five high-level Domains, which range from “acute threat” in the Negative Valence Systems Domain to “self-knowledge” in the Social Processes Domain. Such diversity of function echos the panoply of theories and empirical phenomena for which ACC has become famous (Bush, 2009; Holroyd & Yeung, 2011). Taken at face value, this taxonomy would seem to suggest that serious disruption to ACC processing – say, as would result from an extensive lesion – would upset all of its associated constructs, inducing wholesale impairments across all five of the domains. We suggest that this multidimensional conceptualization of ACC function, which seems to point to a role for ACC in everything, is not useful for advancing research into psychopathology.

Here we offer a recent theory of ACC function as a new construct for understanding the role of ACC in mental disorders. Our argument is based on our previous proposal that ACC

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motivates the execution of extended behaviors according to principles of hierarchical reinforcement learning, which we call the HRL-ACC theory (Holroyd & Yeung, 2012). In keeping with the dimensional approach, we argue that this function is expressed along continua throughout the general population, manifesting as individual differences in personality that relate to reward sensitivity and persistence (Patrick & Hajcak, 2016). Further, we propose that the extreme expression of these ACC-related traits contributes to a variety of psychiatric and neurological disorders including depression, substance abuse, attention-deficit hyperactivity disorder (ADHD), Parkinson's disease, schizophrenia and obsessive-compulsive disorder (OCD). Further, in accord with emerging principles of computational psychiatry (Adams et al., 2015; Maia & Frank, 2011; Montague et al., 2012), we elaborate this proposal within the context of a computational model (Holroyd & McClure, 2015) that elucidates how the disorders interrelate according to shared impairments within a distributed neural system. Empirical support is derived mainly from two electrophysiological phenomena produced in human ACC: the reward positivity, a component of the event-related brain potential (ERP) associated with reward processing (Proudfit, 2015; Sambrook & Goslin, 2015; Walsh & Anderson, 2012), and frontal-midline theta (FMT) power, an electroencephalogram (EEG) oscillation associated with cognitive control and sustained mental effort (Cavanagh & Frank, 2014; Hsieh & Ranganath, 2014; Mitchell, McNaughton, Flanagan, & Kirk, 2008). This conceptual framework suggests that ACC supports a single underlying construct – motivation of extended, effortful behaviors according to principles of reinforcement learning – rather than the array of seemingly unrelated constructs and phenomena as currently conceived (Table 1).

## Anterior Cingulate Cortex Function

ACC is widely understood to contribute to cognitive control and decision-making but its

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specific role within that broad domain is controversial (Bush, 2009; Mars, Sallet, Rushworth, & Yeung, 2011). Current theories span a plethora of functions that include conflict processing, reinforcement learning, motivating behavior, error detection, action selection and more (Holroyd & Yeung, 2011). However, we have recently proposed a theory of ACC that, we believe, unifies many of these disparate perspectives (Holroyd & Yeung, 2012). The theory holds ACC responsible for learning the value of tasks, selecting tasks based on those learned values, and motivating task execution, which it effects by applying control over the activity of action-production systems. On this view, the specific level of control is regulated according to rewards received during the task, such that ACC applies only the minimal control necessary to support adequate task performance (Holroyd & McClure, 2015, Holroyd & Yeung, 2012; Shahnazian & Holroyd, unpublished results).

The theory extends the “actor-critic” computational architecture that has been commonly used to describe the neural mechanisms of decision making and reinforcement learning (Cohen & Frank, 2009; Niv, 2009) (Figure 2). In line with current thinking, the actor module, which is implemented by dorsal striatum and dorsolateral prefrontal cortex, maps stimulus inputs to motor outputs. In parallel, the critic module, which is implemented by orbitofrontal cortex and the ventral striatum, receives stimulus input that includes information about unconditioned reinforcers such as primary rewards, as well as “efference copy” of motor commands issued by the actor (Angel, 1976). On the basis of this information the critic computes the value of ongoing events (O’Doherty et al., 2004; Padoa-Schioppa, 2011), as well as reward prediction error (RPE) signals that indicate whether the values are “better than expected” or “worse than expected”. These signals are then conveyed as phasic increases or decreases from baseline dopamine activity, respectively, to the neural targets of the dopamine system (Schultz, 2013), where they

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are utilized by the actor to improve performance on the task at hand according to principles of reinforcement learning, and by the critic to optimize its predictions of future reward (Sutton & Barto, 1998).

Our extension to this architecture develops previous ideas about the neural mechanisms of hierarchical reinforcement learning (HRL; Botvinick, Niv & Barto, 2009; Holroyd & Coles, 2002). We have proposed that ACC selects the action policy – called an “option” in the language of HRL – directed toward a particular goal. Put another way, the ACC selects the *task* for execution. The actor then performs the individual actions that carry out the task while the critic monitors those actions. On this view, the ACC would be responsible for deciding to go for a long run, directing the actor to place one foot ahead of the other and the critic to monitor progress toward the finish line. The theory also holds that the tasks are selected based on the average value of the rewards received during task execution, as determined from RPE signals conveyed to the ACC from the critic via the midbrain dopamine system (Figure 2; see also Holroyd & Coles, 2002). These reward signals can be elicited by physical rewards, such as food, or by abstract rewards, such as progress toward a goal.

Further, in line with previous theories of ACC function (e.g., Botvinick, Braver, Barch, Carter, & Cohen 2001; see also Shenhav, Botvinick, & Cohen, 2013), the theory proposes that ACC regulates the deployment of control via a feedback-control loop. Crucially, a computational model that implements this control mechanism depends on three important assumptions (Holroyd & McClure, 2015). First, the model assumes that the actor is effort-averse such that it eschews the production of effortful behaviors, especially when these yield relatively smaller rewards, as observed for example in studies of rats with ACC damage (Walton, Bannerman & Rushworth, 2002). For this reason the actor’s actions sometimes decouple from the higher-level

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objective specified by the ACC – as might occur, for example, if the actor balked at running up a steep hill when tasked to do so by the ACC. Second, the model assumes that the ACC control signal attenuates these costs, thereby ensuring that the actor produces behaviors that comport to the selected task. Thus the control signal deployed by the ACC would enable the actor to ignore the cost associated with running up the hill. Third, the model assumes that the control level starts high when a task is initially put into execution, and then applies only the minimal amount of control necessary to maintain a high average reward rate, as determined from the feedback-control loop: control decays when the received rewards are as large or larger than the average reward for that task, and increases otherwise (Holroyd & McClure, 2015; see also Holroyd, 2016). This mechanism enables the system to explore a range of actions in novel task contexts where the degree of effort necessary to maximize reward is unknown, and to gradually withdraw control as the actor learns a successful action policy for that environment. Hence tasks that do not require control can be carried out successfully even when the ACC is impaired, whereas tasks that do require control cannot (Holroyd & Yeung, 2012). The theory also proposes that multiple control modules are hierarchically organized along the rostral-caudal extent of ACC (Holroyd & McClure, 2015), and that recurrent connecting loops between the striatum, thalamus, and ACC enable the ACC to track the actor's progress toward the task's completion (Shahnazian & Holroyd, unpublished results) (see the direct projection from actor to ACC in Figure 2). However, these latter aspects of the theory go beyond the scope of this proposal and are discussed only briefly below.

This conceptualization holds the ACC responsible for regulating effortful behavior in terms of larger task objectives that extend beyond the moment-to-moment pursuit of relatively ephemeral rewards. The theory thus aligns with the observation that motivational arousal

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increases with task difficulty until the reward value of the task is no longer perceived to be worth the effort (Brehm & Self, 1989; Brehm & Wright, 1983). In practice, this process facilitates the completion of effortful tasks associated with low immediate reward value, as effortful behaviors with high immediate reward value can be carried out by the actor independently of the ACC (Holec, Pirot, & Euston, 2014). A substantial body of evidence in support of the HRL-ACC theory has been reviewed elsewhere (Holroyd & McClure, 2015; Holroyd & Yeung, 2011, 2012; Shahnazian & Holroyd, unpublished results; see also Shenhav et al., 2013; Verguts, Vassena, & Silvetti, 2015 for comparable ideas). Further, the model proposes somewhat different roles for rostral ACC (dorsal to the genu of the corpus callosum) and caudal ACC (called “anterior midcingulate cortex” in the nomenclature of Vogt, 2009), which we return to in the discussion section. For the present purpose suffice it to note that, in keeping with the RDoC framework, the theory suggests that individual differences in ACC function should underlie differences in the ability to select and motivate extended, effortful tasks according to the rewards received during task execution, and that mental disorders associated with ACC dysfunction reflect the extreme, maladaptive expression of these traits.

Especially instructive about the consequences of ACC dysfunction are clinical studies in humans that causally manipulate ACC activity. For example, electrical stimulation of ACC has been observed to induce compulsive goal-directed motor behaviors (Kremer, Chassagnon, Hoffmann, Benabid, & Kahane, 2001) and a feeling of perseverance in order to achieve a distant goal (Parvizi, Rangarajan, Shirer, Desai, & Greicius, 2013). As one patient described the feeling, “There are so many millions of small things that happen to you daily that, you have to handle them, you have to deal with them. It’s the major things that if you give up on, you’re in trouble. You can’t give up” (Parvizi et al., 2013). At the other end of this spectrum, widespread damage

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to ACC (Anderson, Arciniegas, Huddle, & Leehey, 2003; Degos, da Fonseca, Gray, & Cesaro, 1993; Devinsky, Morrell, & Vogt, 1995; Le Beau, 1954; Nagaratnam, Nagaratnam, Ng, & Diu, 2004), or a lack of dopamine input to ACC (Németh, Hegedüs, & Molnár, 1988), can cause akinetic mutism, a state characterized by dramatic reductions in spontaneous speech and movement in the presence of preserved motor ability. As was reported of one patient following remission of her illness, "She didn't talk because she had 'nothing to say.' Her mind was 'empty'. Nothing 'mattered'. She apparently was able to follow . . . conversations even during the early period of the illness, but felt no 'will' to reply" (Damasio & Van Hoesen, 1983). Damage to ACC (Naccache et al., 2005) and electrical disruption of the adjacent supplementary motor area (Zenon, Sidibe, & Olivier, 2015) can also extinguish feelings of effortful control.

In between these extremes the sequelae of ACC damage are typically less dramatic. In contrast to the sometimes florid presentation of people with other frontal impairments, it has been said that the person with ACC damage "shows nothing obviously unusual in his behavior when one sees him, meets him or talks to him: to all intents and purposes he is normal" (Tow & Whitty, 1953). Yet closer examination of such patients reveals a subtle array of deficits including reduced drive and less interest in engaging in creative activities (Tow & Whitty, 1953). When assessed in the laboratory, ACC lesions disrupt the ability to sustain a rewarded response across multiple trials in humans (Camille, Tsuchida, & Fellows, 2011; see also Amiez, Champod, Wilson, Procyk, & Petrides, 2015) and in non-human primates (Amiez, Joseph, & Procyk, 2006; Buckley et al., 2009; Chudasama et al., 2013; Kennerley, Walton, Behrens, Buckley, & Rushworth, 2006), evidently because of an impaired ability to maintain attention to the task at hand (Rushworth, Hadland, Gaffan, & Passingham, 2003). Humans with ACC damage also exhibit response slowing and increased variability in speeded response time (RT) tasks

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(Alexander, Stuss, Shallice, Picton, & Gillingham, 2005; Modirrousta & Fellows, 2008; Picton, Stuss, Alexander, et al., 2006; Picton, Stuss, Shallice, Alexander, & Gillingham, 2006; Stuss et al., 2005; Vendrell et al., 1995), giving rise to the hypothesis that ACC is responsible for “motivating” or “energizing” behavior (Alexander et al., 2005; Kouneiher, Charron, & Koechlin, 2009; Picton, Stuss, Alexander, et al., 2006; Picton, Stuss, Shallice, et al., 2006; Stuss et al., 2005; Tucker, Luu, & Pribram, 1995). Even in healthy individuals, trials with high RT are preceded by transient decreases in ACC activity together with greater activation of the so-called default network, which is suggestive of momentary lapses of attention (Weissman, Roberts, Visscher, & Woldorff, 2006). The HRL-ACC theory accounts for such findings by proposing that disruptions to ACC function – whether because of physical damage or normal fluctuations in activation – results in the withdrawal of control over action production; in the absence of the motivational supervision, behaviors become slower and less accurate.

## Electrophysiological Measures of ACC Function

This line of research is complicated by the fact that the sensations of reward valuation and effort expenditure are available only to introspection. Although sometimes utilized as a proxy for the latter, task performance is a poor indicator of effort expenditure: Individuals can do well either because a task is easy, requiring little effort, or because the deployment of control can make a difficult task *appear* to be easy. Motivational factors such as fatigue can also impair task performance, or not, depending on how they are managed (Kool & Botvinick, 2012; Mockel, Beste, & Wascher, 2015; Sarter, Gehring, & Kozak, 2006; Westbrook & Braver, 2015), and effort expenditure is not even always aversive (Csikszentmihalyi, 1990). Attempts to infer subjective reward values from choice behavior are equally problematic (Skinner, 1953). For these reasons, well-characterized neural measures of effortful control and reward valuation can

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provide invaluable, independent evidence of these processes, rendering a host of interesting questions amenable to scientific investigation (Botvinick & Braver, 2015; Padoa-Schioppa, 2011). According to the HRL-ACC theory, these two key processes align in ACC, which utilizes the RPE signals conveyed from the critic (reflecting subjective reward valuation) to assess how much control to allocate over task performance (reflecting subjective effort expenditure) (Figure 2). Crucially, we propose that these factors manifest in two electrophysiological signals generated within ACC: the reward positivity and FMT (HajiHosseini & Holroyd, 2013; Holroyd, HajiHosseini, & Baker, 2012) (Figure 2). If this hypothesis is correct, then these signals provide a window onto the subjective phenomenology of these processes.

FMT describes a band of EEG oscillations in the theta frequency range (4-8 Hz) recorded over frontal-central areas of the scalp. The phenomenon has been investigated for many decades. Consistent with the HRL-ACC theory, a multitude of studies have indicated that FMT power correlates positively with levels of cognitive effort, working memory load and attention, especially for tasks that demand sustained control (Berger, Omer, Minarik, Sterr, & Sauseng 2015; Griesmayr, Gruber, Klimesch, & Sauseng, 2010; Ishihara & Yoshii, 1972; Kornhuber, Lang, Kure, & Kornhuber, 1990; Lang, Lang, Diekmann, & Kornhuber, 1987; Lang, Lang, Kornhuber, Diekmann, & Kornhuber, 1988; Mizuki, Tanaka, Isozaki, Nishijima, & Inanaga, 1980; Nakagawa, 1988; Sasaki et al., 1996; Vogel, Broverman, & Klaiber, 1968; Yamaguchi, 1981; Yamamoto & Matsuoka, 1990; for reviews see Hsieh & Ranganath, 2014; Mitchell et al., 2008). Shorter bursts of theta power that extend for only a few cycles are also associated with momentary increases in cognitive control (Cavanagh & Frank, 2014). Further, converging evidence indicates that FMT is generated in ACC: FMT has been source-localized to ACC (Asada, Fukuda, Tsunoda, Yamaguchi, & Tonoike, 1999; Berger et al., 2015; Gevins, Smith,

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McEvoy, & Yu, 1997; Ishii et al., 1999, 2014; Luu & Tucker, 2001; Nigbur, Ivanova, & Stürmer, 2011; Onton, Delorme, & Makeig, 2005; Sauseng, Hoppe, Klimesch, Gerloff, & Hummel, 2007; Scheeringa et al., 2008), and electrical stimulation of human ACC induces theta oscillations (Talairach et al., 1973). Intracranial recordings in rodents (Feenstra & Holsheimer, 1979; Holsheimer, 1982; Young & McNaughton, 2009), monkeys (Tsujimoto, Shimazu, & Isomura, 2006; Tsujimoto, Shimazu, Isomura, & Sasaki, 2003, 2010) and humans (Cohen, Ridderinkhof, Haupt, Elger, & Fell, 2008; Wang, Ulbert, Schomer, Marinkovic, & Halgren, 2005) also confirm ACC as an important source of theta oscillations.

An accumulating body of evidence indicates that FMT serves to bind distributed neural systems into functional networks for the purpose of task execution (Anderson, Rajagovindan, Ghacibeh, Meador, & Ding, 2010; Benchenane et al., 2010; Clayton, Yeung, & Kadosh, 2015; Cohen, 2011; Daitch et al., 2013; Jones & Wilson, 2005; Klimesch, Freunberger, Sauseng, & Gruber, 2008; Likhtik, Stujenske, Topiwala, Harris, & Gordon, 2014; Mitchell et al., 2008; Mizuhara & Yamaguchi, 2007; Sarnthein, Petsche, Rappelsberger, Shaw, & von Stein, 1998; Sato & Yamaguchi, 2007; Sauseng et al., 2007; Schack, Vath, Petsche, Geissler, & Möller, 2002; Scheeringa et al., 2008; Womelsdorf, Johnston, Vinck, & Everling, 2010; Womelsdorf, Vinck, Leung, & Everling, 2010; Young & McNaughton, 2009). We have argued elsewhere that this neural synchronization, together with other biophysical processes related to the application of control, contributes to increased production and decreased clearance of neural waste products in the interstitial fluid surrounding brain cells (Holroyd, 2016). This hypothesis provides a biophysical rationale for the proposal that ACC applies the minimal degree of control necessary to sustain adequate task performance (Holroyd & McClure, 2015). That FMT can serve as an independent measure of effortful control is suggested by a study in which FMT increased

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linearly with working memory load on a Sternberg memory task for human subjects with high working memory capacity (as assessed using a different memory task), despite FMT not predicting individual performance on the Sternberg task itself (Zakrzewska & Brzezicka, 2014). Gradual increases in FMT related to time-on-task have also been attributed to the effort required to sustain adequate task performance despite increasing levels of cognitive fatigue (Wascher et al., 2014; see also Kirov, Warsawskaya, & Voynov, 1996). Enhanced FMT due to neurofeedback training, which is predicted by midcingulate cortex volume (Enriquez-Geppert et al., 2013), also facilitates performance on tasks that require proactive control (Enriquez-Geppert, Huster, Figge & Hermann, 2014).

By reflecting the control function of ACC, FMT would appear to index ACC output (Figure 2). By contrast, we propose that the reward positivity reflects modulation of this control function by an important input to ACC (Figure 2). The reward positivity is an ERP component that is distributed over frontal-central areas of the scalp, the amplitude of which is differentially modulated by positive and negative feedback stimuli (Miltner, Braun, & Coles, 1997). Converging evidence from source localization studies (Gehring & Willoughby, 2002; Miltner et al., 1997; Santesso, Dillon, et al., 2008), simultaneous recording of EEG/fMRI signals (Becker, Nitsch, Miltner, & Straube, 2014; Hauser, Iannaccone, Ball, et al., 2014) (see also Amiez et al., 2013; Carlson, Foti, Mujica-Parodi, Harmon-Jones, & Hajcak, 2011; Hauser, Iannaccone, Stämpfli, et al., 2014; Procyk et al., 2014), a transcranial direct current stimulation experiment (Reinhart & Woodman, 2014), and intracranial recording studies in rodents (Warren, Hyman, Seamans, & Holroyd, 2015), non-human primates (Emeric et al., 2008) and humans (Smith, E. H., et al., 2015) indicate that the reward positivity is produced in ACC. Originally termed the feedback error-related negativity (Miltner et al., 1997), the ERP component has recently been

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renamed because of its evident sensitivity to rewarding rather than non-rewarding events (Arbel, Goforth, & Donchin, 2013; Baker & Holroyd, 2011; Becker et al., 2014; Eppinger, Kray, Mock, & Mecklinger, 2008; Foti, Weinberg, Dien, & Hajcak, 2011; Holroyd, Krigolson, & Lee, 2011; Holroyd, Pakzad-Vaezi, & Krigolson, 2008; Potts, Martin, Burton, & Montague, 2006; Sambrook & Goslin, 2014; San Martin, Manes, Hurtado, Isla, & Ibañez, 2010; Weinberg, Riesel, & Proudfit, 2014; for a review see Proudfit, 2015).

We have previously argued that the reward positivity reflects the receipt of RPE signals carried by the midbrain dopamine system to ACC, which utilizes the signals for the adaptive modification of behavior according to principles of reinforcement learning (Holroyd & Coles, 2002; for review see Walsh & Anderson, 2012) (Figure 2). Since then, this proposal has found support from several sources of evidence. First, a meta-analysis of 55 data sets, which utilized a novel technique for analyzing published ERP waveforms, has confirmed that reward positivity amplitude behaves as an RPE signal (Sambrook & Goslin, 2015). Second, the signal complies with formal principles of associative learning (Luque, López, Marco-Pallares, Càmar, & Rodríguez-Fornells, 2012) as well as to an axiomatic definition of an RPE signal (Talmi, Fuentemilla, Litvak, Duzel, & Dolan, 2012). Third, the timing of RPE signals recorded directly from the dopamine midbrain nuclei in humans (Zaghloul et al., 2009) coincides with the timing of the reward positivity (Sambrook & Goslin, 2015). Fourth, midbrain grey matter volume in the region of the dopamine nuclei is positively associated with reward positivity amplitude (Carlson, Foti, Harmon-Jones, & Proudfit, 2015). Fifth, individual differences in reward positivity amplitude are modulated by a genetic polymorphism that codes for the expression of dopamine D4 receptors (Baker, Stockwell, Barnes, Haesevoets, & Holroyd, 2016), which are strongly expressed in cortex (Ariano, Wang, Noblett, Larson, & Sibley, 1997; Lauzon & Laviolette, 2010;

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Mrzljak et al., 1996). Sixth, the amplitude of the reward positivity, but not of other ERP components, is selectively blocked by administration of low doses of the dopamine D2/D3 agonist pramipexole, and this effect has been source localized to ACC (Santesso et al., 2009). Seventh, the reward positivity is attenuated in non-human primates by the dopamine antagonist haloperidol (Vezoli & Procyk, 2009). And last, reward positivity amplitude is positively correlated with the fMRI BOLD response in the ventral striatum, underscoring its sensitivity to mesocorticolimbic reward circuit activation (Becker et al., 2014; Carlson et al., 2011; Proudfit, 2015).

Despite the evidence in favor of this proposal, it has been challenged on the basis of mixed findings about whether the reward positivity in fact reflects a reinforcement learning process: Whereas some studies have observed that reward positivity amplitude predicts trial-to-trial changes in behavior (Arbel et al., 2013; Castellar, Kühn, Fias, & Notebaert, 2010; Cohen & Ranganath, 2007; Frank, Woroch, & Curran, 2005; Hewig et al., 2011; Holroyd & Krigolson, 2007; Philiastides, Biele, Vavatzanidis, Kazzner, & Heekeren, 2010; Santesso, Dillon, et al., 2008; van der Helden, Boksem, & Blom, 2010; Yasuda, Sato, Miyawaki, Kumano, & Kuboki, 2004), other studies have failed to do so (Chase, Swainson, Durham, Benham, & Cools, 2011; Luft, Nolte, & Bhattacharya, 2013; Luft, Takase, & Bhattacharya, 2014; Miltner et al., 1997; Sailer, Fischmeister, & Bauer, 2010; van de Vijver, Ridderinkhof, & Cohen, 2011; Walsh & Anderson, 2011; for reviews see Gehring, Liu, Orr, & Carp, 2012; Luft, 2014; Walsh & Anderson, 2012). However, the HRL-ACC theory holds that the simple stimulus-response mappings that mediate such trial-to-trial changes in behavior are learned by the actor in brain areas outside of ACC, i.e., by the actor (Holroyd & Yeung, 2012) (Figure 2). Instead, the ACC is said to learn the reward value of entire task *policies* (Holroyd & Coles 2002; Holroyd & Yeung, 2012). Because the

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theory holds that the dopamine-dependent RPE signals are carried to both the ACC (which produces the reward positivity) and the actor module (which adjusts behavior according to the size of the RPE signals) (Figure 2), reward positivity amplitude is nevertheless expected to be weakly correlated with feedback-based changes to behavior—as in fact appears to be the case.

By learning the value of the task policies, the ACC is not only well-positioned to select the policies on the basis of those learned values, but also to motivate performance of the task itself, which it effects by slowly regulating the degree of control it applies over task execution (Holroyd & McClure, 2015). Consistent with this proposal, the amplitude of the reward positivity in fact appears to reflect an interaction between the tonic level of control applied by ACC over task performance (as indexed by FMT) and the size of the RPE signal that adjusts the control level (Baker, Stockwell, et al., 2016; HajiHosseini & Holroyd, 2013). Further, a small but growing literature suggests that reward positivity amplitude is sensitive to contextual factors that motivate behavior. For example, reward positivity amplitude is determined by the global rather than local reward context in which rewards are delivered (Kujawa, Smith, Luhmann, & Hajcak, 2013), generalizes to the task contexts in which the actions are carried out (Umemoto, HajiHosseini, Yates, & Holroyd, unpublished results), and is larger for task contexts in which participants either win or not as compared to contexts in which they either avoid losses or not (Rongjun, Yuejia, Zheng, & Xiaolin, 2007; Weinberg et al., 2014). Reward positivity amplitude is also affected by state factors: The component is attenuated by a sad mood induction (Foti, Kotov, Klein, & Hajcak, 2011) and amplified to preferred rewards by an anger induction, which has been interpreted as reflecting increased motivation to pursue an otherwise frustrated goal-directed course of action (Angus, Kemkes, Schutter, & Harmon-Jones, 2015). Learning-related increases in reward positivity amplitude have also been said to indicate a clearer assessment of

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task achievement (Arbel, Murphy, & Donchin, 2014).

Finally, the reward positivity shares a functional relationship with another ERP component called the error-related negativity (ERN), which immediately follows error commission in speeded response time tasks (Falkenstein, Hohnsbein, Hoormann, & Blanke, 1990; Gehring, Goss, Coles, Meyer, & Donchin, 1993). Like the reward positivity, the ERN is also generated in the ACC (see Gehring et al., 2012, for review). Because ERN onset coincides with response initiation (Gehring et al., 1993) and has been observed in a completely deafferented patient (Allain, Hasbroucq, Burle, Grapperon, & Vidal, 2004), the mechanism that produces it is believed to depend on efference copy of the motor command (Angel, 1976) (Figure 2). We have proposed that the ERN and the reward positivity constitute different manifestations of a single underlying phenomenon, the former elicited by RPE signals triggered by the response and the latter by RPE signals triggered by feedback stimuli (Holroyd & Coles, 2002). Yet despite this apparent functional commonality, in practice the ERN and reward positivity have been examined mostly in separate literatures. Our review therefore focuses on the reward positivity and highlights the ERN findings only where these are especially relevant.

## ACC and Personality

The RDoC approach characterizes the symptomatology of mental disorders in terms of underlying constructs that express dimensionally across the population. On this view, variability in the neural implementation of these constructs gives rise to individual differences in behavior that, in extreme cases, manifest in specific constellations of symptoms. This perspective suggests that if the ACC motivates the selection and execution of extended behaviors, as we have argued, then that function should be expressed differently from one person to the next. Put another way, ACC function may underlie one or more personality traits associated with the motivation of

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extended behaviors.

The HRL-ACC model highlights two key computational processes related to ACC input and output that would be expected to contribute to individual differences in behavior (Figure 2). On the output side of the equation, ACC may contribute to personality traits related to effortful control, such as persistence (Cloninger, Svrakic, & Przybeck, 1993) and cognitive control (Tangney, Baumeister, & Boone, 2004), as well as to traits associated with their absence like apathy (Clarke et al., 2011; Marin, Biedrzycki, & Firinciogullari, 1991) and impulsivity (Whiteside & Lynam, 2001). Anxiety may also be related to ACC function, given that anxious people appear to devote extra processing resources to task performance (Eysenck & Calvo, 1992). On the input side of this equation, ACC activity would be expected to contribute to personality traits related to reward processing like reward responsiveness (Van den Berg, Franken, & Muris, 2010), as well as to the lack thereof such as anhedonia (Chapman, Chapman, & Raulin, 1976; Snaith, Hamilton, Morley, Humayan, Hargreaves, & Trigwell, 1995). In view of the model depicted in Figure 2, we would expect that sensitivity to particular sorts of rewards would tilt the control function of ACC toward favoring tasks that elicit such rewards. ACC function should contribute especially to traits that reflect commitment to extended tasks that demand high investments in effort, especially when the immediate reward value for doing so is low.

Recent studies have implicated ACC function in many of the above personality traits, especially for tasks that entail careful assessments of effort expenditure and reward value. For example, apathy – the hallmark of akinetic mutism resulting from ACC damage (Levy & Dubois, 2006; Stuss, van Reekum, & Murphy, 2000; van Reekum, Stuss, & Ostrander, 2005) – is negatively correlated in Alzheimer's disease with ACC grey matter volume (Apostolova et al.,

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2007; Stanton, Leigh, Howard, Barker, & Brown, 2013) and white matter integrity (Hahn et al., 2013; Ota, Sato, Nakata, Arima, & Uno, 2011; Tighe et al., 2012; see also Hollock et al., 2015), and is specifically associated with ACC hypofunction in this population (Migneco et al., 2001). Apathy is also seen to increase with normal aging in healthy individuals (Brodaty, Altendorf, Withall, & Sachdev, 2010; Clarke, Ko, Lyketsos, Rebok, & Eaton, 2010), manifesting as a decreased willingness to exert effort for smaller rewards (Bonnelle, Veromann, et al., 2015). These apparent associations between ACC, effort/reward processing and apathy have recently been confirmed in a landmark study by Bonnelle and colleagues, who found that functional and structural connectivity between ACC and other brain areas was inversely correlated with apathy levels in healthy individuals making effortful choices ( $R^2 = 31\%$  functional,  $R^2 = 25\%$  structural, Bonelle, Manohar, Behrens, & Husain, 2015). Meanwhile, individuals who score high in state and trait anhedonia, which is associated with decreased EEG and fMRI indicators of ACC activation ( $R^2 = 18\%$ , Harvey, Armony, Malla, & Lepage, 2010;  $R^2 = 17\%$ , Mies et al., 2013;  $R^2 = 19\%$  Broadman area 24,  $R^2 = 29\%$  Broadman area 25, Wacker, Dillon, & Pizzagalli, 2009), are less willing to exert effort for larger rewards – especially when successful performance for effortful behaviors is randomly rewarded (Treadway, Bossaller, Shelton, & Zald, 2012; Treadway, Buckholtz, Schwartzman, Lambert, & Zald, 2009). And persistence is associated with an increased fMRI BOLD response in ACC ( $R^2 = 62\%$ , Gusnard et al., 2003), especially when subjects reject a low-effort choice in favor of a more effortful (but more rewarding) behavior (Kurniawan et al., 2010).

Given that the reward positivity and FMT are generated in ACC, these ERP components should be especially revealing as regards to the contribution of ACC to individual differences in personality. Crucially, as befits a measure of a stable personality trait, the test-retest reliability of

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reward positivity amplitude ( $r = .64$  and  $.67$ , Bress, Meyer, & Proudfit, 2015;  $r = .62$ , Schmidt, Holroyd, Debener, & Hewig, 2016;  $r = .71$ , Segalowitz et al., 2010) and FMT ( $r = .85-.90$ , Gold, Fachner, & Erkkilä, 2013) is high. And in fact, reward positivity amplitude is positively correlated with personality traits related to reward sensitivity ( $r = -.33$ , where increasing negative values indicate larger amplitude, Bress & Hajcak, 2013;  $r = .39$ , Cooper, Duke, Pickering, & Smillie, 2014) and with the extroversion-related traits of approach and agency ( $r = .36$ , Cooper et al., 2014;  $r = .46$ , Smillie, Cooper, & Pickering, 2011). Although other studies have yielded mixed findings on this issue, these studies utilized either different methods for measuring reward positivity amplitude, or alternative scales for assessing reward sensitivity (Balconi & Crivelli, 2010; Cooper et al., 2014; Lange, Leue, & Beauducel, 2012; Padrão, Mallorquí, Cucurell, Marco-Pallares, & Rodriguez-Fornells, 2013; Santesso, Dzyundzyak, & Segalowitz, 2011; Van den Berg, Franken, & Muris, 2011). In contrast to the reward positivity, the relationship between FMT and personality has been relatively less explored. When measured relative to theta power over posterior areas of the scalp, low FMT power is – perhaps surprisingly – associated with agentic extroversion, a persistence-like trait that encompasses drive, assertiveness, enthusiasm, dominance, and ambition (Chavanon, Wacker, & Stemmer, 2011; Mueller, Panitz, Pizzagalli, Hermann, & Wacker, 2015). More in keeping with the HRL-ACC theory, personality traits associated with persistence and task engagement predict high ERN amplitude (Tops & Boksem, 2010).

A meta-analysis of FMT studies has also indicated that anxious individuals generate relatively more FMT compared to less anxious individuals (Cavanagh & Shackman, 2015), in line with the proposal that anxious people allocate excessive levels of control over task performance (Eysenck & Calvo, 1992). Anxiety (Hajcak, 2012; Hajcak, McDonald, & Simons,

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2003), neuroticism (Pailing & Segalowitz, 2004) and negative emotionality (Luu, Collins, & Tucker, 2000) are also associated with a larger ERN. Further, whereas the reward positivity to reward feedback stimuli is normal in anxious individuals (Bress, Meyer, & Hajcak, 2015), the reward positivity elicited by *ambiguous* feedback stimuli is relatively small (because the ERPs to negative and ambiguous feedback stimuli are similar) in people who are neurotic (Hirsh & Inzlicht, 2008) or anxious (Gu, Ge, Jiang, & Luo, 2010; see also Gu, Huang, & Luo, 2010) – consistent with the proposal that such individuals find ambiguous feedback to be highly aversive. The negative ERP deflection elicited by error feedback (as opposed to the reward positivity, which is assessed as the difference in the size of the ERPs to positive and negative feedback stimuli; Proudfit, 2015) also appears to be larger in individuals who score high in personality traits related to negative affect (De Pascalis, Varriale, & D’Antuono, 2010; Santesso et al., 2012; Sato et al., 2005). Collectively, these observations converge with a variety of studies that indicate that ACC is especially sensitive to negative affect and to punishment signals (DeYoung et al., 2010; Hajcak, 2012; Shackman et al., 2011).

Computational simulations based on the HRL-ACC theory illustrate how, for environments that afford both small, easily-attained rewards and larger rewards that demand greater investments of effort, the withdrawal of control by ACC can unleash seemingly impulsive behaviors (Holroyd & McClure, 2015). These simulations also reveal that this form of impulsivity is distinguishable from the type of impulsiveness associated with temporal discounting of future rewards (Holroyd & McClure, 2015), which is known to depend strongly on orbitofrontal cortex function (e.g., Rudebeck, Walton, Smyth, Bannerman, & Rushworth, 2006). We defer a discussion of this issue until the end of this proposal. For the present purpose suffice it to note that individuals who highly discount the value of future rewards relative to

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immediate rewards produce, relative to other individuals, larger reward positivities to the immediate rewards, suggesting that they may be more motivated to work for those rewards (Cherniawsky & Holroyd, 2013; see also Massar, Rossi, Schutter, & Kenemans, 2012). Further, recent evidence indicates that the ability to delay gratification, as reflected in reward positivity amplitude, is mediated by personality traits associated with high cognitive control and low impulsivity (Schmidt et al., 2016). Impulsive individuals who are poor planners and prone to making risky decisions also produce relatively larger reward positivities (Onoda, Abe, & Yamaguchi, 2010; cf. Kamarajan et al., 2009, 2010).

## ACC and Psychopathology

Our review suggests that ACC function is related to a spectrum of personality traits that include persistence, reward sensitivity, cognitive control, anxiety, and many others. Unsurprisingly, several of these traits are also associated with a variety of mental and neurological disorders. Here we explore whether these characteristics link ACC dysfunction with six disorders in particular: depression, ADHD, schizophrenia, Parkinson's disease, substance abuse, and OCD. Our analysis is organized around three computational considerations that follow from the HRL-ACC model (Figure 2). First, because ACC selects and motivates task execution based on the learned reward values for these tasks, the withdrawal of control by this system should result in decreased motivation to carry out extended, goal directed behaviors. Second, over-activation of ACC should give rise to increased concern over the successful completion of these extended behaviors. Third, abnormal reward input to ACC should steer control over extended behaviors toward the attainment of particular goals and away from others.

*Depression*

A substantial body of evidence implicates ACC dysfunction in depression (Holtzheimer

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& Mayberg, 2009; Vogt, Fountoulakis, et al., 2009). The HRL-ACC theory predicts that ACC dysfunction should disrupt the ability to utilize rewards for motivating and sustaining effortful control over extended, goal-directed behaviors. As we have argued, these qualities should relate to personality traits associated with ACC function, such as apathy and reward sensitivity, which in turn should manifest in abnormal markers of ACC function including individual differences in FMT power and reward positivity amplitude.

In fact, depression is strongly associated with individual differences in both reward sensitivity and effortful control. With regard to the former, depressed individuals are impaired at adjusting their behavior in response to negative (Elliott, Sahakian, Herrod, Robbins, & Paykel, 1997) and positive (Pizzagalli, Iosifescu, Hallett, Ratner, & Fava, 2009) feedback stimuli, especially when these individuals exhibit elevated levels of anhedonia (Pizzagalli, Jahn, & O'Shea, 2005; Steele, Kumar, & Ebmeier, 2007). The hemodynamic response in ACC is also blunted to RPEs in depressed individuals (Kumar et al., 2008), and to feedback stimuli in depressed people who are highly anhedonic (Mies et al., 2013; Steele et al., 2007; see also Harvey et al., 2010). These abnormal RPEs may reflect abnormal reward prediction, abnormal reward learning, or both, associated with altered ACC activation during reward anticipation (Gorka, Huggins, Fitzgerald, Nelson, Phan, & Shankman, 2014; Knutson, Bhanji, Cooney, Atlas & Gotlib, 2008; Smoski et al., 2009; Smoski, Rittenberg, & Dichter, 2011). Consonant with these observations, Greg Hajcak and colleagues have demonstrated across a series of studies that reward positivity amplitude can serve as a biomarker for depression (see Proudfit, 2015 for review). Reduced reward positivity amplitude is associated with elevated depressive symptoms in children (Bress, Smith, Foti, Klein, & Hajcak, 2012) and adults (Foti & Hajcak, 2009), and with major depressive disorder (Liu et al., 2014), especially when it occurs with impaired mood

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reactivity to positive events (Foti, Carlson, Sauder, & Proudfit, 2014) (cf. Santesso, Steele, et al., 2008; Tucker, Luu, Frishkoff, Quiring, & Poulsen, 2003). Reward positivity amplitude also predicts subsequent onset of first-episode depression in adolescent girls (Bress, Foti, Kotov, Klein, & Hajcak, 2013; Nelson, Perlman, Klein, Kotov, & Hajcak, 2016), and is not associated with symptoms of anxiety, indicating specificity to depression (Bress et al., 2012; Bress, Meyer, & Hajcak, 2015).

An extensive line of research has also demonstrated that the production of effortful behaviors is curtailed in depression (Cohen, Lohr, Paul, & Boland, 2001; Hartlage, Alloy, Vázquez, & Dykman, 1993; Zakzanis, Leach, & Kaplan, 1998; but see Hammar et al., 2011), evidently because of an impaired ability of rewards to motivate effortful behavior (Sherdell, Waugh, & Gotlib, 2012; Treadway et al., 2012), especially when effort is needed to obtain large rewards (Barch, Treadway, & Schoen, 2015; Clery-Melin et al., 2011; Treadway et al., 2012). Apathy in depression is associated with decreased ACC grey matter volume, suggesting that this characteristic may stem partly from ACC dysfunction (Lavretsky, Ballmaier, Pham, Toga, & Kumar, 2007). Neuroimaging studies have also revealed decreased resting-state activation of ACC (Pizzagalli, Peccoralo, Davidson, & Cohen, 2006), as well as reduced ACC activation in complex tasks that demand cognitive effort (Elliott, Baker, et al., 1997)—unless performance is equated across groups, in which case depressed individuals show increased ACC activation (Harvey et al., 2005). Furthermore, we have argued elsewhere that the sensation of cognitive effort stems in part from the role of ACC in synchronizing the activity of other brain areas with FMT (Holroyd, 2016). This proposal accords with the suggestion that depression compromises the ability of ACC to act as the central hub of the default network (Pizzagalli, 2011), and with an observation that FMT power is positively correlated with ACC glucose metabolism in normal,

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but not depressed, individuals (Pizzagalli, Oakes, & Davidson, 2003). Although another study failed to find a relationship between resting-state FMT and depression (Gold et al., 2013), the HRL-ACC theory suggests that individual differences in FMT should relate to performance on tasks that demand sustained control, not to the resting state per se.

Collectively, these studies are consistent with the proposal that depression disrupts the ability by ACC to motivate control over extended behaviors in accord with the reward value of task execution.

*ADHD*

Childhood ADHD is characterized by a tendency to engage in task-irrelevant behaviors in lieu of the task at hand. Nevertheless, children with ADHD are commonly observed to sustain attention on tasks that are intrinsically rewarding (Bush, 2010; Geurts, Luman, & van Meel, 2008). The HRL-ACC theory points to the ACC-dopamine interface, which directs control to tasks with high reward value (Holroyd & McClure, 2015; Holroyd & Yeung, 2012), as the source of this effect. On our account, this aspect of ACC function should express across the general population as a dimension of personality. And in fact, adults with ADHD score low on the “Big 5” personality trait of conscientiousness, which reflects the ability to accomplish projects in an efficient, self-organized way (Nigg et al., 2002), and which is strongly correlated with the trait of persistence (Umemoto & Holroyd, in press).

Several sources of evidence implicate dopamine and ACC function in the etiology of ADHD. In particular, a large literature indicates that reward processing and reinforcement learning are abnormal in children with ADHD (Douglas & Parry, 1983, 1994; Haenlein & Caul, 1987; Parry & Douglas, 1983; Wender, 1972; see Luman, Oosterlaan, & Sergeant, 2005 for review). Not unlike the consequences of ACC damage in humans (Camille et al., 2011) and non-

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human primates (Amiez et al., 2006; Buckley et al., 2009; Chudasama et al., 2013; Kennerley et al., 2006), the choice behavior of children with ADHD is relatively more sensitive to the most recent reward obtained than to previous reward history (Tripp & Alsop, 1999; see also Kollins, Lane, & Shapiro, 1997; Luman, Oosterlaan, Knol, & Sergeant, 2008). Relative to typically developing children, they also exhibit stronger preferences for immediate rewards (Tripp & Alsop, 2001), elevated temporal discounting of rewards (Scheres, Tontsch, Thoeny, & Kaczurkin, 2010), and progressive increases in risky decision-making (Drechsler, Rizzo, & Steinhausen, 2008). The heart rate (Crone, Jennings, & van der Molen, 2003) and skin conductance response (Masunami, Okazaki, & Maekawa, 2009) of children with ADHD are also relatively less sensitive to differences between reward and penalty feedback.

These behaviors appear to be propelled by an exaggerated sensitivity to the salience of rewards (Carlson & Tamm, 2000; Luman et al., 2005; McInerney & Kerns, 2003; Scheres, Lee, & Sumiya, 2008), such that simply providing external motivators like money, tokens, and social rewards tends to normalize differences in cognitive control (Dovis, Van der Oord, Wiers, & Prins, 2012; Kohls, Herpertz-Dahlmann, & Konrad, 2009; Konrad, Gauggel, Manz, & Schöll, 2000; Shiels et al., 2008). Their behavior can also be normalized by contextual effects in which relatively small rewards are delivered before larger rewards (Huang-Pollack, Mikami, Pfiffner, & McBurnett, 2007) and by increasing the rate of reinforcement (Aase & Sagvolden, 2006; but see Barber, Milich, & Welsch, 1996; Luman, van Meel, Oosterlaan, Sergeant, & Geurts, 2009). Likewise, although children with ADHD are averse to waiting during delays (Sonuga-Barke, Taylor, Sembi, & Smith, 1992), this aversion can be ameliorated by providing rewarding stimuli like simple coloured cartoons during the waiting period (Antrop et al., 2006). Atypical sensitivity to salient rewards is also revealed by the reward positivity, which reaches adult size in typically

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developing children as young as 10 years old (Lukie, Montazer-Hojat, & Holroyd, 2014), but in children with ADHD is highly dependent on the reward value of the feedback: Unlike the case for typically developing children, for these individuals feedback stimuli indicating immediate monetary rewards elicits a larger reward positivity than feedback stimuli indicating abstract points (Holroyd, Baker, Kerns, & Müller, 2008; Umemoto, Lukie, Kerns, Mueller, & Holroyd, 2014; cf. Rosch & Hawk, 2013; van Meel, Heslenfeld, Oosterlaan, Luman, & Sergeant, 2011).

An equally substantial literature implicates midbrain dopamine system abnormalities in ADHD (see Tripp & Wickens, 2009, for a review). A natural inference is that disrupted dopamine-dependent RPE signals are responsible for the observed reward processing deficits in ADHD, which has spawned a range of reinforcement learning-based theories of the disorder (e.g., Grace, 2001; Johansen et al., 2009; Sagvolden, Johansen, Aase, & Russell, 2005; Silvetti, Wiersema, Sonuga-Barke, & Verguts, 2013; Tripp & Wickens, 2008; Williams, 2008; Williams & Taylor, 2004; see Luman, Tripp, & Scheres, 2010 for a review). A formal model comparison has suggested that many of these observations can be accounted for by positive RPE signals that are asymmetrically larger than the (absolute) size of negative RPE signals (Cockburn & Holroyd, 2010).

Further, ACC appears to be the neural locus in ADHD that links abnormal dopamine reward signals to control over extended behaviors (Bush, 2009, 2010). ACC is relatively hypoactive in this population (Durstun et al., 2007), especially for tasks that demand cognitive control (Konrad, Neufang, Hanisch, Fink, & Herpertz-Dahlmann, 2006; Rubia et al., 2009, 2010). Studies that compare the reward-positivity with the fMRI BOLD response also implicate abnormal ACC activation in this reward processing deficit (Hauser, Iannaccone, Ball, et al., 2014). ACC also produces less evoked FMT (Groom et al., 2010) and is less functionally

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connected with posterior brain areas (Castellanos et al., 2008) in this population. In line with the diminished “energizing” effect on RT following ACC damage in adults (Alexander et al., 2005; Modirrousta & Fellows, 2008; Picton, Stuss, Alexander, et al., 2006; Picton, Stuss, Shallice, et al., 2006; Stuss et al., 2005; Vendrell et al., 1995), the RTs of children with ADHD tend to be slower and more variable relative to those of typically developing children (Castellanos, Sonuga-Barke, Milham, & Tannock, 2006; Johnson et al., 2007). It has thus been proposed that measures of ACC function can constitute an endophenotype or intermediate phenotype of ADHD (Albrecht et al., 2008; Durston, de Zeeuw, & Staal, 2009; Tye, McLoughlin, Kuntsi, & Asherson, 2011).

Stimulant medications may correct the system by boosting the reward value of tasks via artificially elevating dopamine levels. Administration of psychostimulants such as methylphenidate, which increase extrasynaptic dopamine levels both subcortically (Cragg & Rice, 2004) and in prefrontal cortex (Arnsten, 2006; Berridge et al., 2006), improves the behavioral and cognitive symptoms of ADHD (de Sonnaville, Njioiktjen, & Bos, 1994; DeVito et al., 2008; Rubia et al., 2009; Tannock, Schachar, Carr, Chajczyk, & Logan, 1989), a change that appears to be mediated by increased ACC activation (Bush et al., 2008; Bush, 2009; Epstein et al., 2007; Rubia et al., 2009). Remarkably, even reduced ACC grey matter volume in ADHD (Carmona et al., 2005; Seidman et al., 2006) is normalized by treatment with stimulant medications (Pliszka, Lancaster, Liotti, & Semrud-Clikeman, 2006; Semrud-Clikeman, Pliszka, Bledsoe, & Lancaster, 2012). Such long-term structural changes have been observed in non-human animal studies, where repeated self-administration of amphetamine leads to increased pyramidal cell spine density in medial prefrontal cortex (Crombag, Gorny, Li, Kolb, & Robinson, 2005), resulting in larger and more sustained activity of these neurons (Homayoun &

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Moghaddam, 2006).

*Schizophrenia*

The first meeting on Cognitive Neuroscience for Treatment Research to Improve Cognition in Schizophrenia – a precursor to the NIMH RDoC working groups – identified two cognitive control-related functions that appear to be impaired in schizophrenia: rule generation and selection, and dynamic adjustments in control following conflicts, errors and other events (Kerns, Nuechterlein, Braver, & Barch, 2008). Yet the negative symptoms of schizophrenia, which are characterized by reductions in movement, cognitive activity, and emotion, point instead to impaired motivation of goal-directed behaviors (Brown & Pluck, 2000; Kring & Barch, 2014). The HRL-ACC theory holds that this function is mediated by the dopamine-ACC interface, especially for extended, effortful tasks, and thus that disruption to this system might contribute to the negative symptoms of schizophrenia.

In fact, schizophrenia is strongly associated with disturbances to ACC function, dopamine activity, and their interaction. Problems with ACC include structural and functional abnormalities (Benes, Cunningham, Berretta, & Gisabella, 2009; Preda, Rilling, Chin, & Tamminga, 2009) such as reduced grey matter volume (Mitelman, Shihabuddin, Brickman, Hazlett, & Buchsbaum, 2005) and altered cytoarchitecture and circuitry (Benes et al., 2009; Benes, Majocha, Bird, & Marotta, 1987; Benes, Vincent, Alsterberg, Bird, & SanGiovanni, 1992). Functional neuroimaging (Carter, MacDonald, Ross, & Stenger, 2001; Carter, Mintun, Nichols, & Cohen, 1997; Laurens, Ngan, Bates, Kiehl, & Liddle, 2003; Preda et al., 2009) and ERP (Mulert et al., 2001) studies also indicate decreased ACC activation to task events. Moreover, the dominant biologically-based theory of the disorder points to abnormal dopamine levels (Howes & Kapur, 2009), which appear to upset ACC processing in schizophrenia: reduced

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concentrations of the dopamine metabolite 3,4-dihydroxyphenylacetic acid are found in ACC in this population (Wyatt, Karoum, & Casanova, 1995), and administration of apomorphine – a dopamine agonist – normalizes ACC function, as revealed by positron emission tomography (Dolan et al., 1995).

On the face of it, these deficits suggest disrupted reward valuation in schizophrenia much as they do in depression, and indeed a long-standing hypothesis has been that the impairments in goal-driven behavior observed in people with schizophrenia are due to anhedonia (Kring & Barch, 2014). Yet recent investigations have indicated that people with schizophrenia actually value rewards normally; for example, they report normal levels of positive emotion when presented with rewarding stimuli, and develop normal implicit reward biases in reward learning tasks (Gold, Waltz, Prentice, Morris, & Heerey, 2008; Heerey, Bell-Warren, & Gold, 2008; Kring & Barch, 2014). Consistent with these observations – and in stark contrast to both people with depression and children with ADHD – people with schizophrenia generate a normal reward positivity across a range of tasks (Morris, Holroyd, Mann-Wrobel, & Gold, 2011; see also Krawitz, Braver, Barch, & Brown, 2011; Morris, Heerey, Gold, & Holroyd, 2008), as well as a normal fMRI response to positive and negative feedback stimuli (Paulus, Frank, Brown, & Braff, 2003).

If not anhedonia, then what causes the negative symptoms of schizophrenia? A telling clue comes from ERP studies that have consistently revealed smaller ERN amplitudes in this population (Alain, McNeely, He, Christensen, & West, 2002; Bates, Kiehl, Laurens, & Liddle, 2002; Bates, Liddle, Kiehl, & Ngan, 2004; Kopp & Rist, 1999; Mathalon et al., 2002; Morris et al., 2008, 2011; Morris, Yee, & Nuechterlein, 2006), as well as a reduced BOLD response to errors in ACC and brainstem (e.g., Polli et al., 2008). These observations suggest that people

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with schizophrenia have trouble associating reward values (which are normal) to the actions that motivate the pursuit of those rewards (which are abnormal), in line with behavioral (Gold et al., 2008, 2012), electrophysiological (Morris et al., 2011) and functional neuroimaging evidence (Morris, Quail, Griffiths, Green, & Balleine, 2015) to this effect. Transcranial direct-current stimulation over the medial-frontal cortex also normalizes ERN amplitude in people with schizophrenia, but mainly for those who express fewer delusion-related positive symptoms; the stimulation is ineffective for people with severe negative symptoms (Reinhart, Zhu, Park, & Woodman, 2015), suggesting that the ability of ACC to support extended behaviors may be more seriously impaired in this population.

In contrast to the evidence that people with schizophrenia value rewards normally, several recent studies have converged on the finding that the negative symptoms of schizophrenia reflect hypersensitivity to effort-related costs of reward acquisition (Barch et al., 2014; Culbreth, Westbrook & Barch, 2016; Fervaha, Graff-Guerrero, Zakzanis, Foussias, Agid, & Remington, 2013; Gold, J. M., et al., 2013; Hartmann et al., 2015; Strauss et al., in press; Wolf et al., 2014; but see Gold et al., 2015). This impairment may result from ACC dysfunction. As observed for people with ACC damage and for children with ADHD (see above), the response times of people with schizophrenia are usually slower and more variable than those of control subjects; ERP studies indicate that this slowness results from a problem with response selection rather than stimulus processing (Luck et al., 2009). A recent neuroimaging study has also associated the negative symptoms of schizophrenia with abnormal *over-activation* of ACC when rewards are utilized to motivate and sustain behavior, which was interpreted by the investigators as a compensatory response (Park et al., 2015). In another fMRI study, the ACC BOLD signal in control subjects revealed an interaction between working memory load and the reward value of

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the task, yet the direction of this activation profile was reversed for people who scored highly on a personality factor related to the negative symptoms of schizophrenia (termed “diminished expression”) (Hager et al., 2015). Notably, the HRL-ACC theory also predicts that, to the extent that low-level actions are effortful, impairments to ACC function should impede exploration of the task domain (Holroyd & McClure, 2015), and in fact the negative symptoms of schizophrenia have been associated with reduced exploratory behavior (Strauss et al., 2011).

Taken together, these studies implicate impaired ACC processing in the negative symptoms of schizophrenia, resulting in reduced motivation to carry out extended, goal-directed behaviors despite normal reward valuation.

*Substance Abuse*

The cardinal feature of substance abuse – continued drug consumption despite long-term adverse consequences (Lubman, Yücel, & Pantelis, 2004) – is indicative of abnormal decision making and cognitive control (Herrnstein, 1997; Potenza, 2007; Rahman, Sahakian, Cardinal, Rogers, & Robbins, 2001). Substance abusers make relatively risky decisions (Lane & Cherek, 2000; Rogers et al., 1999), discount the value of future drug rewards more than they do future monetary rewards (Madden, Bickel, & Jacobs, 1999), and are generally insensitive to the delayed consequences of their actions (Petry, Bickel, & Arnett, 1998). Impulsivity is also a vulnerability factor for substance dependence (Dawe & Loxton, 2004; Sher, Bartholow, & Wood, 2000; Verdejo-Garcia, Lawrence, & Clark, 2008). On the face of it, this personality trait might be assumed to reflect the withdrawal of control by ACC over brain areas that are relatively more sensitive to immediate rewards (Figure 2). Impulsivity is in fact positively correlated with reward positivity amplitude (Schmidt et al., 2016), which suggests that the ACC/dopamine interface may contribute to an aspect of impulsivity underlying substance dependence. Yet ACC

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damage typically leads to apathy rather than impulsivity, so it would be an oversimplification to suggest that the contribution of ACC to substance dependence simply reflects a withdrawal of control.

The link between dopamine and substance abuse is especially strong (Wise, 1996). All addictive drugs stimulate dopamine release from midbrain dopamine neurons (Cami & Farre, 2003; Di Chiara & Imperato, 1988; Koob & Le Moal, 1997; Marinelli, Cooper, Baker, & White, 2003; Rice & Cragg, 2004; Samochowiec, Kleinrok, & Horodnicki, 1995; Tang & Dani, 2009; Willuhn, Burgeno, Groblewski, & Phillips, 2014), which carry the RPE signals used for reinforcement learning (Montague, Hyman, & Cohen, 2004; Schultz, 2013). This characteristic has given rise to the hypothesis that drug dependence is fundamentally a problem of learning and memory: By potentiating the function of the dopamine system, addictive substances act to reinforce maladaptive drug-taking behaviors (Di Chiara, 1999; Hyman, 2005). On this view, drugs of abuse increase the size of phasic dopamine signals by raising extracellular dopamine levels (Grace, 2000), effectively increasing the magnitude of positive RPEs. In turn, the augmented RPEs cause the motivational value of states that precede drug consumption to increase without bound, biasing the action-selection mechanism to favor behaviors that ultimately converge on drug use (Redish, 2004). These changes are mediated by synaptic modification at the molecular level (Berke & Hyman, 2000; Hyman & Malenka, 2001) that result in the rewiring of underlying neural networks (Robinson & Kolb, 1997, 1999; Robinson, Gorny, Mitton, & Kolb, 2001; for review see Robinson & Kolb, 2004).

Although much of this theoretical and empirical work has focused on how these changes impact the striatum, drugs of abuse also disrupt cortical function (Volkow, Wang, Fowler, & Tomasi, 2012; Volkow, Wang, Fowler, Tomasi, & Telang, 2011). ACC dysfunction is

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particularly implicated in substance dependence (Peoples, 2002): gray matter volume in ACC and other cortical areas (Franklin et al., 2002), dopamine D2 receptor availability in ACC (Volkow et al., 1993), ACC activation (Forman et al., 2004; Garavan & Hester, 2007; Kaufman, Ross, Stein, & Garavan, 2003), and ERN amplitude (Franken, van Strien, Franzek, & van de Wetering, 2007) are all reduced in stimulant abusers. The HRL-ACC theory suggests that drug-potentiated dopamine signals should bias individuals to engage in extended behaviors that lead to drug consumption, thereby associating these behaviors with enhanced reward value while de-potentiating the value of other, non-drug-related activities. Support for this proposal comes from three experiments indicating that people who abuse addictive substances produce a relatively small reward positivity to nominal monetary incentives (Baker, Stockwell, et al., 2016; Baker, Stockwell, Barnes, & Holroyd, 2011; Baker, Wood, & Holroyd, 2016; see also Baker & Holroyd, 2015; Potts, Bloom, Evans, & Drobles, 2014), but following an overnight period of abstinence, the size of the reward positivity is normalized by feedback stimuli that signify a chance to puff on a cigarette (Baker et al., 2016). Problem gamblers also produce especially large reward positivities to gambling rewards (Hewig et al., 2010; cf. Oberg, Christie, & Tata, 2011). These observations indicate that the reduced response to so-called natural rewards in addicted populations does not reflect a global deficit in reward processing.

Moreover, the impairment appears to be mediated by the dopamine-ACC interface, as reward positivity amplitude and FMT power statistically mediate the impact on substance dependence of a genetic polymorphism that codes for the expression of dopamine D4 receptors (Baker, Stockwell et al., 2016). D4 receptors are strongly expressed in cortex (Ariano et al., 1997; Lauzon & Laviolette, 2010; Mrzljak et al., 1996) and are said to play a prominent role in disturbed emotional processing underlying substance abuse and other neuropsychiatric disorders

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(Lauzon & Laviolette, 2010). For example, mice lacking this receptor are hypersensitive to ethanol, cocaine and methamphetamine, but are generally hypoactive (Rubinstein et al., 1997), perhaps reflecting diminished task engagement by ACC to non-drug related activities. Excessive drug use is also associated with decreased sensitivity to natural rewards as revealed in behavior (Bolanos, Barrot, Berton, Wallace-Black, & Nestler, 2003; Koob & Le Moal, 1997, 2005), the hemodynamic response in ACC (Martin-Soelch et al., 2001; see also Wilson et al., 2014), and dopamine processing in ACC and other cortical areas (Volkow, Fowler, Wang, & Goldstein, 2002; Volkow, Wang, Fowler, Baler, & Telang, 2009; see also Goldstein et al., 2007). By contrast, the ACC is relatively *more* activated when cocaine-abusers are exposed to stimuli that induce cravings (Volkow et al., 2011), as would be expected if ACC were energized by such stimuli to initiate pursuit of the drug.

Finally, in one study, the ACC of both control and cocaine-addicted participants became less activated with increasing fatigue due to time-on-task, yet only the control subjects showed a concomitant increase in the size of midbrain dopamine error signals; administration of methylphenidate drove the midbrain dopamine response in opposite directions across the two groups, which the authors interpreted as evidence that drugs of abuse compromise the midbrain dopamine system's ability to sustain effortful control over behavior (Moeller, Tomasi, Honorio, Volkow, & Goldstein, 2012). Also suggestive is an observation that the ACC is activated by cognitive reappraisal (a cognitive strategy in which individuals reinterpret the meaning of an emotional situation) of the opportunity to smoke on a cigarette (Zhao et al., 2012).

Our analysis suggests that impulsivity in addiction does not reflect a lack of control *per se*; rather, it reflects the maladaptive redirection of control. The reward positivity thus appears to reflect the consequences of drugs of abuse as they “usurp” the control system (Hyman, 2007).

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*Parkinson's Disease*

Parkinson's disease results from progressive deterioration of the midbrain dopamine nuclei (Lotharius & Brundin, 2002) that precipitates a variety of motor impairments that are the signature characteristic of the disorder (Pavese, 2012). Yet for decades investigators have also recognized a variety of cognitive deficits in the disease that are indicative of impaired frontal cortical function (Au et al., 2012; Cools, 2006; Cools, van den Bercken, Horstink, van Spaendonck, & Berger, 1984; Downes et al., 1989; Levin, Llabre, & Weiner, 1989; Morris et al., 1988; Owen et al., 1995; Taylor, Saint-Cyr, & Lang, 1986, 1990; Wright, Burns, Geffen, & Geffen, 1990), as well as atypicalities in reinforcement learning (Frank, Seeberger, & O'Reilly, 2004; Knowlton, Mangels, & Squire, 1996; Vriezen & Moscovitch, 1990). Several studies have also observed functional (e.g., Grossman, Crino, Reivich, Stem, & Hurtig, 1992; Playford et al., 1992; Samuel et al., 1997; Vogt, Vogt, Purhoit, & Hof, 2009) and structural (Kovari et al., 2003) abnormalities in ACC. Because the disease primarily affects the dopamine projections to the striatum, impairments to higher cortical function are often assumed to occur indirectly via interconnecting striatal-thalamo-cortical pathways. Nevertheless, Parkinson's disease can also affect dopamine neurons that project directly to ACC. As stated in one review, "It has never been made clear that the dopamine dysfunction hypothesis applies to anterior midcingulate cortex; not just the substantia nigra and its projections," an oversight that is attributed to technical considerations (Vogt, Vogt, et al., 2009, p. 718). Does ACC dysfunction in fact contribute to the symptoms of Parkinson's disease, and is this a consequence of reduced dopamine signals in ACC?

The HRL-ACC theory suggests that wholesale dysfunction of ACC in Parkinson's disease should result in impaired effortful control over extended behaviors. At odds with this

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proposal is the observation that patients with Parkinson's disease score low in the personality trait of novelty seeking (Menza, Golbe, Cody, & Forman, 1993) and can therefore be said to be persistent, among other characteristics associated with this trait (Cloninger, 1986). Yet consistent with the theory, people with Parkinson's disease are especially impaired at executing everyday tasks that require sustained cognitive effort (Weingartner, Burns, Diebel, & LeWitt, 1984). They also exert fewer willed behaviors due to heightened sensitivity to the energetic costs of movement (Mazzoni, Hristova, & Krakauer, 2007; Niv & Rivlin-Etzion, 2007), and abandon cognitive tasks relatively early when encountering problems of a certain level of difficulty (Schneider, 2007). RT slowing is also specifically associated with frontal lobe dysfunction in Parkinson's disease (Berry, Nicolson, Foster, Behrmann, & Sagar, 1999), and Parkinson's disease with depression, which often co-occur (Starkstein et al., 1989), is associated with diminished ACC activity (Mentis et al., 2002; Ring et al., 1994) that may reflect structural changes to the ACC bundle (Kostić & Filippi, 2011). Further, as compared to non-depressed individuals with Parkinson's disease, depressed people with Parkinson's disease exhibit lower levels of an *in vivo* marker of a dopamine and norepinephrine transporter binding in ACC, suggesting that depression in Parkinson's disease results from loss of catecholaminergic projections to ACC (Remy, Doder, Lees, Turjanski, & Brooks, 2005). Loss of dopaminergic input to human ACC also promotes cognitive impairments associated with Parkinson's disease with dementia (Vogt, Vogt, et al., 2009).

Finally, apathy in Parkinson's disease, which can occur independently of depression (Czernecki et al., 2002; Starkstein et al., 1992), is associated with between 15% to 70% of cases; the high variance in these estimates is evidently due to the different diagnostic approaches employed (Pagonabarraga, Kulisevsky, Strafella, & Krack, 2015). Reports of mental and

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physical fatigue by patients, which are believed to reflect high levels of apathy (Fabbrini et al., 2013; Pagonabarraga et al., 2015), are also associated with ACC dysfunction (Pagonabarraga et al., 2015) and improve following administration of dopaminergic medication (Czernecki et al., 2002). Apathy and depression can also result from dopamine denervation of ACC and other cortical and limbic areas as a consequence of surgical interventions for Parkinson's disease (Thobois et al., 2010). Tellingly, reward positivity amplitude is reduced in apathetic people with Parkinson's disease, as compared both to non-apathetic people with Parkinson's disease and to controls subjects, thus specifically implicating impaired reward processing by ACC in people with Parkinson's disease who are apathetic (Martinez-Horta et al., 2014).

Our analysis suggests that symptoms of apathy and depression in Parkinson's disease may partly result from diminished dopaminergic input to ACC, leading to decreased motivation for carrying out extended behaviors.

*OCD*

OCD is characterized by intrusive thought patterns and ritualized behaviors that are aversive and distressful to the individual (Hezel & McNally, in press). Given the HRL-ACC model, it is tempting to speculate that whereas apathy reflects decreased task engagement by ACC, the symptoms of OCD reflect hyper-concern by ACC over task execution. On this view, ACC continuously works to suppress possible – but highly unlikely – mental and behavioral acts that would otherwise interfere with progress toward a superordinate goal. Thus, for example, obsessive worry over whether or not the front door was locked after leaving home in the morning could interfere with the larger objective of traveling to work. These considerations suggest that we should find evidence of abnormal ACC function in OCD.

In fact, OCD is associated with numerous structural and functional abnormalities in ACC

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(see Piras, Piras, Caltagirone, & Spalletta, 2013; Saxena, O'Neill, & Rauch, 2009 for reviews). For example, OCD patients demonstrate cortical thinning (Kuhn, Kaufmann, Simon, Endrass, Gallinat & Kathmann, 2013; Norman, Carlisi, Lukito, Hart, Mataix-Cols, Radua & Rubia, 2016; Tang, Zhu, Gong, Zhu, Wang, & Chen, 2016), neurochemical abnormalities (Gnanavel, Sharan, Khandelwal, Sharma & Jagannathan, 2014; O'Neill et al., 2016 ) and reduced dopamine receptor binding (Olver et al., 2010) in ACC. ACC is hyper-activated in OCD (Perani et al., 1995), especially for tasks that demand high levels of cognitive control (Fitzgerald et al., 2005; Huyser, Veltman, Wolters, de Haan, & Boer, 2010; Saxena, O'Neill, et al., 2009). Functional neuroimaging studies also suggest that ACC fails to switch to new tasks following task completion (Gu et al., 2008; Remijnse et al., 2013). FMT (Koprivová et al., 2011; cf. Min, Kim, Park, & Park, 2011) and ERN amplitude (Gehring, Himle, & Nisenson, 2000; Johannes et al., 2001; Mathews, Perez, Delucchi, & Mathalon, 2012; Riesel, Endrass, Kaufmann, & Kathmann, 2011) are also elevated in OCD, yet reward positivity amplitude appears to be normal (Endrass, Koehne, Riesel, & Kathmann, 2013; Nieuwenhuis, Nielen, Mol, Hajcak, & Veltman, 2005; Schuller et al., 2015; for review see Endrass & Ullsperger, 2014). Tellingly, ACC has been (Levin & Duchowny, 1991) and continues to be (Banks et al., 2015) a psychosurgery locus for OCD, with ACC grey matter volume and connectivity distinguishing responders and non-responders to the surgery, which indicates that ACC plays a causal role in OCD symptomatology (Banks et al., 2015). A systematic review of 89 structural and functional neuroimaging and electrophysiology studies in OCD suggests that mis-specified cognitive control signals by ACC, which drive the pursuit of tasks that do not comport to long-term goals, constitute a central feature of the disorder (McGovern & Sheth, 2016).

In view of the HRL-ACC model, these observations suggest that ACC contributes to

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OCD by applying excessive control over the execution of extended tasks, as well as by possibly failing to switch to different tasks following task completion.

## Discussion

The RDoC framework is inspiring a broad-based effort to identify common neural origins of a variety of seemingly disparate mental disorders (e.g., Dillon et al., 2014; Griffiths, Morris, & Balleine, 2014; Weinberg, Dieterich, & Riesel, in press; Woody & Gibb, 2015). Yet this search is complicated by the fact that many of the units in the RDoC matrix are associated with multiple domains and constructs. For example, the ERN has been linked with Positive Valence Systems, Negative Valence Systems, and Cognitive Systems (Baskin-Sommers & Foti, in press; see also Olvet & Hajcak, 2008; Weinberg, Riesel, & Hajcak, 2012); given the multiplicity of mappings between the ERN and its associated constructs, how can ERN amplitude be utilized to make useful inferences about cognitive function? Obfuscating matters further, the ACC-related units in the RDoC matrix have been associated with all five of the high-level domains (Table 1). Fortunately, the methods of computational psychiatry hold out promise for resolving such complexity (Adams et al., 2015; Maia & Frank, 2011; Montague et al., 2012). Here we have embraced this perspective by providing a theoretical formalism that relates the symptoms of six different disorders to ACC dysfunction. If this formulation is correct, then the spectrum of constructs attributed within the RDoC framework to ACC (Table 1) may in fact reflect a single underlying construct concerned with the selection and motivation of extended behaviors. On this view, different sorts of impairments to this function give rise to a signature collection of symptoms that cuts across multiple psychopathologies (Figure 1, bottom).

Of course, ACC constitutes only one element within a larger network (Figure 2), which raises the question: Do abnormal indicators of ACC activity reflect intrinsic impairments to the

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ACC itself, or abnormal input to ACC from other brain areas? Structural abnormalities in ACC clearly point to local dysfunction, but atypical patterns of ACC activity are more challenging to interpret. For instance, reduced reward positivity amplitude in Parkinson's disease with apathy (Martinez-Horta et al., 2014) could indicate malfunction of ACC proper (where this ERP component is generated) or reduced ventral striatal activation (which is strongly correlated with reward positivity amplitude; Becker et al., 2014; Carlson et al., 2011). ACC-related impairments in effortful control could also reflect basal ganglia dysfunction (e.g., Adam et al., 2013; Le Heron et al., 2015; Levy & Dubois, 2006). In what follows, we utilize the model illustrated in Figure 2 as conceptual scaffolding for a discussion about ACC's contribution to the six disorders reviewed above. The particulars of this discussion are necessarily speculative, but an attempt to integrate these disparate sources of evidence within a single theoretical framework would seem to be warranted.

Although the reward positivity is produced in ACC, it is powered by value signals that are computed in the ventral striatum and orbitofrontal cortex, which are believed to implement the critic in the actor-critic reinforcement learning architecture (Holroyd & Coles, 2002; Holroyd & Yeung, 2012; O'Doherty et al., 2004; Padoa-Schioppa, 2011). Any disturbance to the reward processing function of these brain areas would therefore be communicated to the ACC in the form of abnormal RPE signals (Figure 2). For example, reduced reward positivity amplitude in depression could reflect a fundamental disturbance in reward processing by the ventral striatum (Proudfit, 2015). Yet people with depression provide typical affective ratings for primary rewards and punishments, which suggests that the critic in this population processes unconditioned reinforcers like sweetness and pain normally ("reward input" in Figure 2). Rather, depressed individuals appear less able than other people to develop abstract, "model-based"

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predictions of reward value (Adams et al., 2015), which are proposed to be computed in orbitofrontal cortex and then transmitted via the ventral striatum to the midbrain dopamine system (McDannald et al., 2012). Thus, reduced reward positivity amplitude in depression could indirectly reflect abnormal, model-based reward processing by orbitofrontal cortex (Figure 2, circle a).

To return to the example of Parkinson's disease: Reduced ventral striatal metabolism in this population prior to surgery for subthalamic nucleus deep brain stimulation predicts increased post-operative risk of apathy (Robert et al., 2014). Therefore the smaller reward positivity in people with Parkinson's disease who are apathetic (Martinez-Horta et al., 2014) could in fact result from decreased ventral striatal activation. On the other hand, the cardinal feature of Parkinson's disease is deterioration of midbrain dopamine neurons, including the dopamine projection to ACC (Remy et al., 2005; Thobois et al., 2010; Vogt, Vogt, et al., 2009). Thus a more parsimonious explanation would be that the smaller reward positivity in Parkinson's disease with apathy results from attenuated RPE signals consequent to mesocortical dopamine neuron damage (Figure 2, circle b). In either case, the HRL-ACC theory predicts that the diminished dopamine input would cause the ACC to disengage from carrying out extended, goal-directed behaviors – the signature characteristic of apathy.

By contrast, the negative symptoms of schizophrenia include many of the same problems that attend both depression and Parkinson's disease – apathy in particular – but reward positivity amplitude is normal in this population (Morris et al., 2011). The contrast between these diseases is therefore especially telling: Unlike the case with the other disorders, the normal reward positivity in schizophrenia indicates that these symptoms *do not* result from abnormal RPE input to ACC (Figure 2, dashed line). Rather, the symptoms must arise from another source. From

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whence, then, do they spring?

An important clue can be found in theories of schizophrenia that link the disorder to impaired processing of efference copy (Feinberg, 1978; Frith, Blakemore, & Wolpert, 2000; Frith & Done, 1988, 1989; Malenka, Angel, Hampton, & Berger, 1982; Malenka, Angel, Thiemann, Weitz, & Berger, 1986). According to this view, a breakdown in the ability to monitor efference copy in schizophrenia gives rise to confusion about the source of conscious thoughts and actions (Feinberg, 1978; Frith & Done, 1988). Many of these studies have emphasized a connection to positive symptoms (Blakemore, Wolpert, & Frith, 2002; Frith & Done, 1988; Haggard, Martin, Taylor-Clarke, Jeannerod, & Franck, 2003), but one experiment has implicated abnormal processing of efference copy in avolition and apathy in schizophrenia (Ford, Roach, Faustman, & Mathalon, 2008). Schizophrenia is also strongly associated with smaller ERN amplitudes (Alain et al., 2002; Bates et al., 2002, 2004; Kopp & Rist, 1999; Mathalon et al., 2002; Morris et al., 2006, 2008, 2011), which are believed to depend on an efference copy of the erroneous motor command (Allain et al., 2004; Gehring et al., 1993); and computational simulations of the abnormal ERN that take into account normal reward positivity amplitude and normal performance on trial-and-error learning tasks point to disrupted processing along the pathway that relays efference copy from the actor to the critic (Morris et al., 2011) (Figure 2, circle c).

On the basis of these considerations we speculate that the negative symptoms of schizophrenia arise from a parallel disturbance to efference copy sent from the actor to ACC (Figure 2, circle d). A number of recent findings are broadly consistent with this supposition. First, transcranial direct-current stimulation over the medial–frontal cortex normalizes ERN amplitude for people who exhibit severe positive symptoms of schizophrenia, but not for people

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with severe negative symptoms (Reinhart et al., 2015), suggesting that impaired processing of efference copy by ACC reflects a core, unyielding factor underlying the negative symptoms. Second, in a case study, ACC damage specifically impaired the ability to utilize efference copy to guide sequence learning (Amiez et al., 2015). And third, recent computational simulations of sequence execution illustrate how the ACC can utilize efference copy to monitor task progression in the service of HRL (Shahnazian & Holroyd, unpublished results). Abnormal efference copy in schizophrenia would therefore impair the ability of ACC to chain individual actions together into an overall action policy that supports effortful, goal-driven behavior.

Although reward positivity amplitude is also normal in OCD, the ERN is larger than normal in OCD and smaller than normal in schizophrenia (Gehring et al., 2000; Johannes et al., 2001; Mathews et al., 2012; for review see Endrass & Ullsperger, 2014). ACC dysfunction in OCD might therefore be diametrically opposite to that in schizophrenia, indicating excessive concern over the task at hand. Consistent with this possibility, OCD and related anxiety disorders are associated with high FMT, which is argued to reflect exaggerated levels of control (Cavanagh & Shackman, 2015). The source of this problem is not entirely clear, but it could indicate intrinsic ACC dysfunction (Figure 2, circle e), or alternatively, abnormal input to ACC from other neural systems (not shown in Figure 2).

Finally, in contrast to schizophrenia and OCD (which are characterized by relatively normal reward positivity amplitudes), and depression and Parkinson's disease (which are characterized by smaller reward positivity amplitudes), ADHD and substance abuse are associated with reward positivities that are neither larger nor smaller per se. To be sure, as compared to control subjects both of these populations produce smaller reward positivities for certain types of rewards: for children with ADHD the ERP component is reduced to feedback

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indicating abstract points, and for people who abuse addictive substances it is reduced to feedback indicating modest financial gains. Yet these signals are normalized for both populations by feedback stimuli indicating larger or more meaningful rewards: physical money in the case of the former, and opportunities to smoke a cigarette in the case of the latter (Baker et al. 2016; Umemoto et al., 2014). ADHD and addiction are also highly comorbid in adults (Downey, Stelson, Pomerleau, & Giordani, 1997; Kessler et al., 2006; Schubiner et al., 1995), which suggests that such uneven reward valuation may reflect a common underlying construct. We propose that this commonality may arise partly from the impact of abnormal dopamine RPE signals on ACC. In the case of ADHD, computational simulations suggest that positive RPE signals are asymmetrically larger than negative RPE signals, possibly because of abnormal reward valuation in the critic (Cockburn & Holroyd, 2010), in particular the ventral striatum (Scheres, Milham, Knutson, & Castellanos, 2007) (Figure 2, circle f). Similarly, the critic could devalue natural rewards relative to drug rewards in substance dependence, an asymmetry that would be compounded by artificial inflation of positive prediction errors by the drugs themselves (Redish, 2004) (Figure 2, circle g). In both cases, the impact of these skewed RPE signals on ACC would steer the deployment of control toward the execution of maladaptive tasks.

Often the consequences of this abnormal reward valuation would be relatively benign. When called upon to execute an onerous, non-rewarding task, control over the performance of that task should gradually wane, leading to its lackadaisical execution and/or to a switch to a more engaging activity (Holroyd & McClure, 2015). Thus, instead of attending to a boring homework assignment, a child with ADHD might stay up late into the night engrossed in a good novel. This conceptualization resonates with Barkley's description of child development as the emergence of complex, hierarchically organized goal-directed actions that span longer and

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longer time intervals, serving to shift behavior away from a reliance on immediate rewards (Barkley, 1997). We have proposed that this shift reflects the gradual development of options by ACC with long-term reward value (Lukie et al., 2014), a developmental process that may be slowed in children with ADHD (Barkley, 1997). Our proposal is likewise consistent with Sergeant, Oosterlaan, and van der Meere (1999) “energetics” theory of ADHD, which attributes the symptoms of ADHD to diminished effort expenditure – as would transpire following developmental delays in the production of options used to support effortful behavior.

More ominously, altered dopamine reward signals due to drug use would hijack the ACC, inducing a shift in control toward the execution of extended behaviors concerned with drug acquisition (Baker et al., 2011; Baker, Stockwell, et al., 2011). In line with this proposal, intracranial recordings in rats indicate that the nucleus accumbens and medial prefrontal cortex together code for action sequences that lead to cocaine self-administration (Chang, Janak, & Woodward, 1997; Chang, Sawyer, Paris, Kirillov, & Woodward, 2000). This conceptual framework also suggests a possible source of relapse in addiction. Abstinence in drug addiction can be characterized as an ongoing battle between an actor oriented toward drug consumption (due to the dopamine-potentiating effects of chronic substance abuse), and an ACC that applies control over the actor in order to enforce compliance with action policies that are incompatible with that behavior. Successful abstinence would also entail reprogramming the model-based component of the critic – namely, orbitofrontal cortex (Lucantonio, Stalnaker, Shaham, Niv, & Schoenbaum, 2012) – to redefine abstinence as a high-level goal with intrinsic reward value. Suggestive in this regard are observations that the depletion of control boosts the value of forbidden rewards as represented by orbitofrontal cortex (Wagner, Altman, Boswell, Kelley, & Heatherton, 2013). Sadly, however, the HRL-ACC model predicts that successful task

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performance – in this case, continued abstinence – will result in gradual declining levels of control (Holroyd & McClure, 2015). Under normal circumstances most people can easily rebound from the consequences of waning control, as small action slips boost control back to an adaptive level. But for addicted individuals, the impact of a small drop in control could be disastrous: A single drink at a party might spin out of hand, eliciting a burst of dopamine that would reorient both the actor and ACC toward the pursuit of even more of that reward.

## Conclusion

Many aspects of this proposal require further development. For example, with the exception of methylphenidate, we have not described how medication status interacts with ACC function. Neither have we discussed how various subdivisions within ACC differentially contribute to these disorders (Vogt, 2009), not because these distinctions are unimportant, but rather because we view our proposal as a first-order approximation to a more general theory of the role of ACC in psychopathology. That said, the HRL-ACC theory in fact relates different control functions to different regions of the rostral-caudal extent of ACC, with the more rostral areas exercising control over the more caudal areas (Holroyd & McClure, 2015). As a consequence, the rostral areas would be expected to play a more prominent role in behaviors with even longer temporal horizons, as well as with regulating switches between different tasks that belong to the domain of the more extended task, consistent with the effects of rostral ACC damage on switching behavior in humans (Gläscher et al., 2012) and rats (Ragozzino, Detrick, & Kesner, 1999). These considerations suggest that rostral and caudal ACC should support different aspects of personality (Umemoto & Holroyd, in press), but a full discussion of this issue extends beyond the scope of this proposal.

Finally, as an empirical matter, novel behavioral tasks that are diagnostic of ACC

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dysfunction should be developed (Holroyd & Yeung, 2012; Vogt, Vogt, et al., 2009). The challenge is that such tasks need to be extended, hierarchically organized, and effort-demanding, which are inherently difficult properties to implement in a laboratory setting. As a step in this direction, we have shown that selectively rewarding correct performance of one task over another task in a task-switching paradigm yields a pattern of switch costs (the difference in performance when switching between tasks compared to repeating the same task) indicative of increased control over the rewarded task (Umemoto & Holroyd, 2015). The HRL-ACC theory points to ACC as the interface linking this reward information to the application of control, suggesting that ACC dysfunction should disrupt this pattern of switch costs. In other work, computational simulations suggest a role for ACC in motivating the maintenance of information needed to execute extended action sequences (Shahnazian & Holroyd, 2016); disruption to ACC should therefore disrupt the production of such sequences.

Despite these complexities, the HRL-ACC theory affords a theoretical framework for understanding the neural origin of a variety of symptoms that cut across mental and neurological conditions. The theory suggests that these disorders might be treated, in part, by normalizing the control signal issued from ACC to the actor – whether by increasing it in the case of depression, Parkinson’s disease and schizophrenia, decreasing it in OCD, or redirecting it toward more adaptive behaviors in ADHD and substance abuse. The nature of each treatment plan would necessarily depend on the specifics of that disorder, in some cases demanding a synergistic combination of behavioral, pharmaceutical, and neural interventions. For example, we predict that by normalizing ACC function, the administration of dopamine medication (which ameliorates apathy in people with Parkinson’s disease; Czernecki et al., 2002) would increase the size of the reward positivity (which is otherwise attenuated in this population; Martinez-Horta et

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al., 2014). Likewise, the theory predicts that treatment gains for OCD resulting from cognitive behavioral therapy (McKay et al., 2015) result from the downregulation of ACC function, as is in fact suggested by functional neuroimaging studies (Huyser et al., 2010; O'Neil et al., 2013; Saxena, Gorbis, et al., 2009). This conceptual framework may therefore facilitate the development of novel therapeutic approaches that treat the causes of mental disorders rather than their symptoms.

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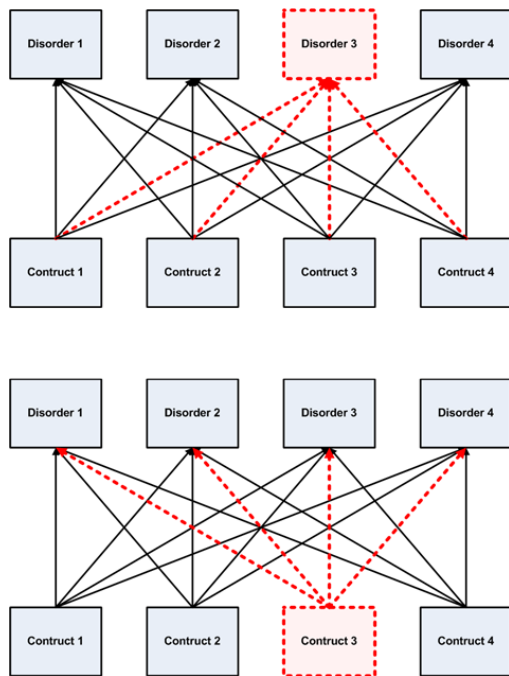
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## Figure Captions

**Figure 1.** Schematic contrasting conventional and Research Domain Criterion (RDoC)

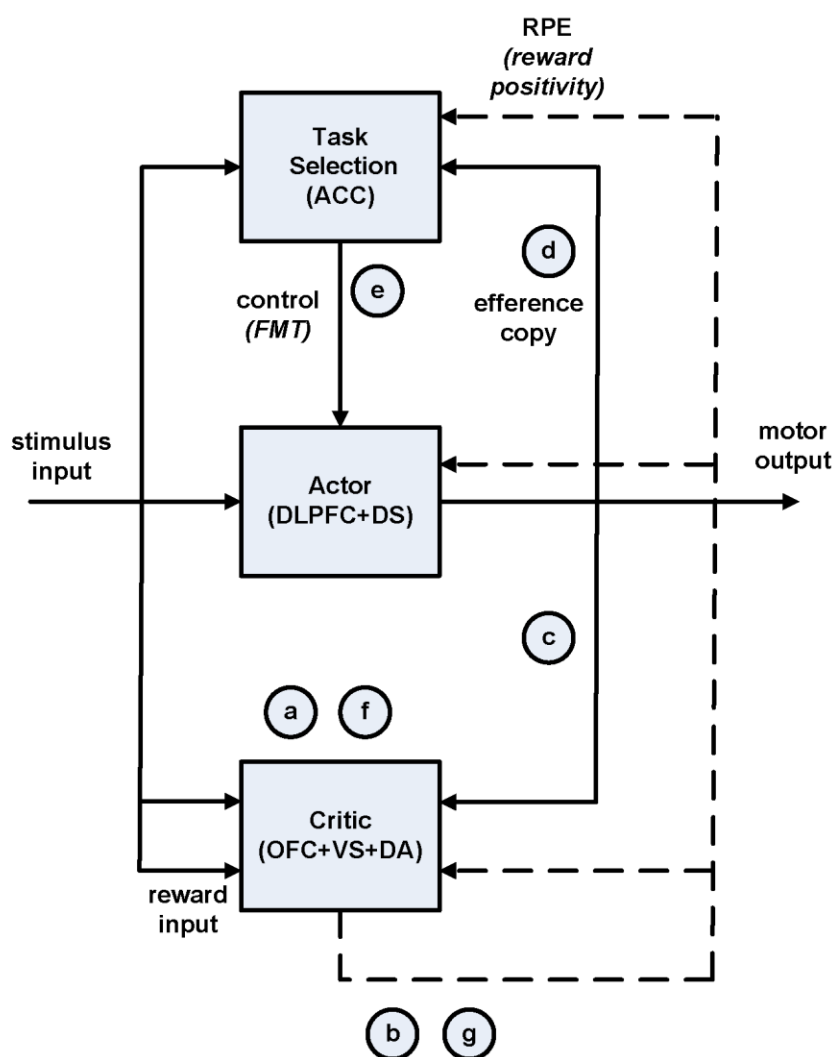
approaches. Each arrow represents an individual symptom or collection of symptoms. Top panel: According to conventional approaches for diagnosing mental disorders, each disorder is defined according to a cluster of symptoms (dashed red). Bottom Panel: By contrast, the RDoC approach focuses on understanding the disorders at the level of the underlying construct, each of which is mediated by a particular, distributed neural system. On this view, symptoms resulting from disruption to a particular construct can contribute to the etiology of multiple disorders (dashed red).



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Figure 2. Hierarchical reinforcement learning theory of anterior cingulate cortex function (HRL-ACC theory). The schematic illustrates the proposed role of anterior cingulate cortex (ACC) in selecting and motivating the execution of extended behaviors, i.e., tasks. Solid arrows indicate directions of information flow. Dashed lines indicate projections of the midbrain dopamine system. Circles indicate brain areas proposed to contribute to ACC dysfunction in various mental and neurological disorders: a) depression is associated with attenuated reward valuation in the critic; b) apathy in Parkinson's disease may reflect attenuated dopamine RPEs; c,d) schizophrenia is associated with disrupted efference copy; e) obsessive-compulsive disorder is associated with the over-application of control by ACC; f) attention-deficit hyperactivity disorder may be associated with abnormal reward valuation by the critic; and g) substance abuse is associated with potentiated RPEs to drug rewards. Not depicted are connections from the ACC to the actor and the critic that specify the task-specific action policy (Holroyd & Yeung, 2012). Other aspects of the model not illustrated include multiple hierarchically-organized modules along the rostral-caudal extent of ACC (Holroyd & McClure, 2015), and recurrent loops between the actor and ACC that enable the ACC to track the execution of each task (Shahnazian & Holroyd, 2016). DLPFC: dorsolateral prefrontal cortex. DS: dorsal striatum. OFC: orbitofrontal cortex. VS: ventral striatum. DA: dopamine. FMT: frontal-midline theta. RPE: Reward prediction error.

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Table

Table 1. Domains, constructs and subconstructs currently attributed to anterior cingulate cortex according to the Research

Domain Criteria framework.

| Domain                         | Construct                                   | Subconstruct                         |
|--------------------------------|---------------------------------------------|--------------------------------------|
| Negative Valence Systems       | Acute Threat                                |                                      |
| Negative Valence Systems       | Sustained Threat                            |                                      |
| Negative Valence Systems       | Loss                                        |                                      |
| Positive Valence Systems       | Approach Motivation                         | Effort Valuation/Willingness to Work |
| Positive Valence Systems       | Approach Motivation                         | Expectancy/Reward Prediction Error   |
| Positive Valence Systems       | Initial Responsiveness to Reward Attainment |                                      |
| Cognitive System               | Performance Monitoring                      |                                      |
| Cognitive System               | Active Maintenance                          |                                      |
| Cognitive System               | Flexible Updating                           |                                      |
| Cognitive System               | Limited Capacity                            |                                      |
| Cognitive System               | Interference Control                        |                                      |
| Social Processes               | Reception of Facial Communication           |                                      |
| Social Processes               | Self-knowledge                              |                                      |
| Arousal and Regularoty Systems | Arousal                                     |                                      |